

Chapter 1A: Introduction to Artificial Intelligence

AMCS2104 Fundamentals of Artificial Intelligence

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- What is AI?
- Different Definitions of AI
- The Turing Test Approach
- The Cognitive Modelling Approach
- The Laws of thought Approach
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What is AI?



Some Definitions of AI

"The exciting new effort to make computers think... machines with minds, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998) "AI... is concerned with intelligent behaviour in artifacts." (Nilsson, 1998)

Defining AI

- Most of the definitions can be distinguished into the following 2 dimensions:
 - Human performance vs. rationality
 - Thought processes and reasoning (thinking) vs. behavior (action)

	Human Performance	Rationality
Thought Processes	Thinking Humanly	Thinking Rationally
Behavior	Acting Humanly	Acting Rationally

Defining AI

Thinking Humanly

"The exciting new effort to make *computers think... machines with minds*, in the full and literal sense."
(Haugeland, 1985)

"[The automation of] activities that we associate with *human thinking*, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)

Acting Humanly

"The art of creating machines that *perform functions* that require intelligence when *performed by people*."
(Kurzweil, 1990)

"The study of how to make computers *do things* at which, at the moment, *people are better*."
(Rich and Knight, 1991)

Thinking Rationally

"The study of *mental faculties* through the use of *computational models*."
(Charniak and McDermott, 1985)

"The study of the *computations* that make it possible to perceive, *reason*, and act."
(Winston, 1992)

Acting Rationally

"*Computational Intelligence* is the study of the design of *intelligent agents*."
(Poole et al., 1998)

"AI... is concerned with *intelligent behaviour* in artifacts." (Nilsson, 1998)

Defining AI

- 4 different **approaches** are used to **develop AI systems**:

- **Acting Humanly**

- Known as the **Turing Test approach**.
 - AI systems developed with this approach are capable of acting like human.

- **Thinking Humanly:**

- Known as the **Cognitive Modelling approach**.
 - AI systems developed with this approach are capable of thinking like human.

Defining AI

- 4 different **approaches** are used to **develop AI systems**:

- **Thinking Rationally**

- Known as the "**Laws of Thought**" approach.
 - AI systems developed with this approach are capable of thinking rationally.

- **Acting Rationally**

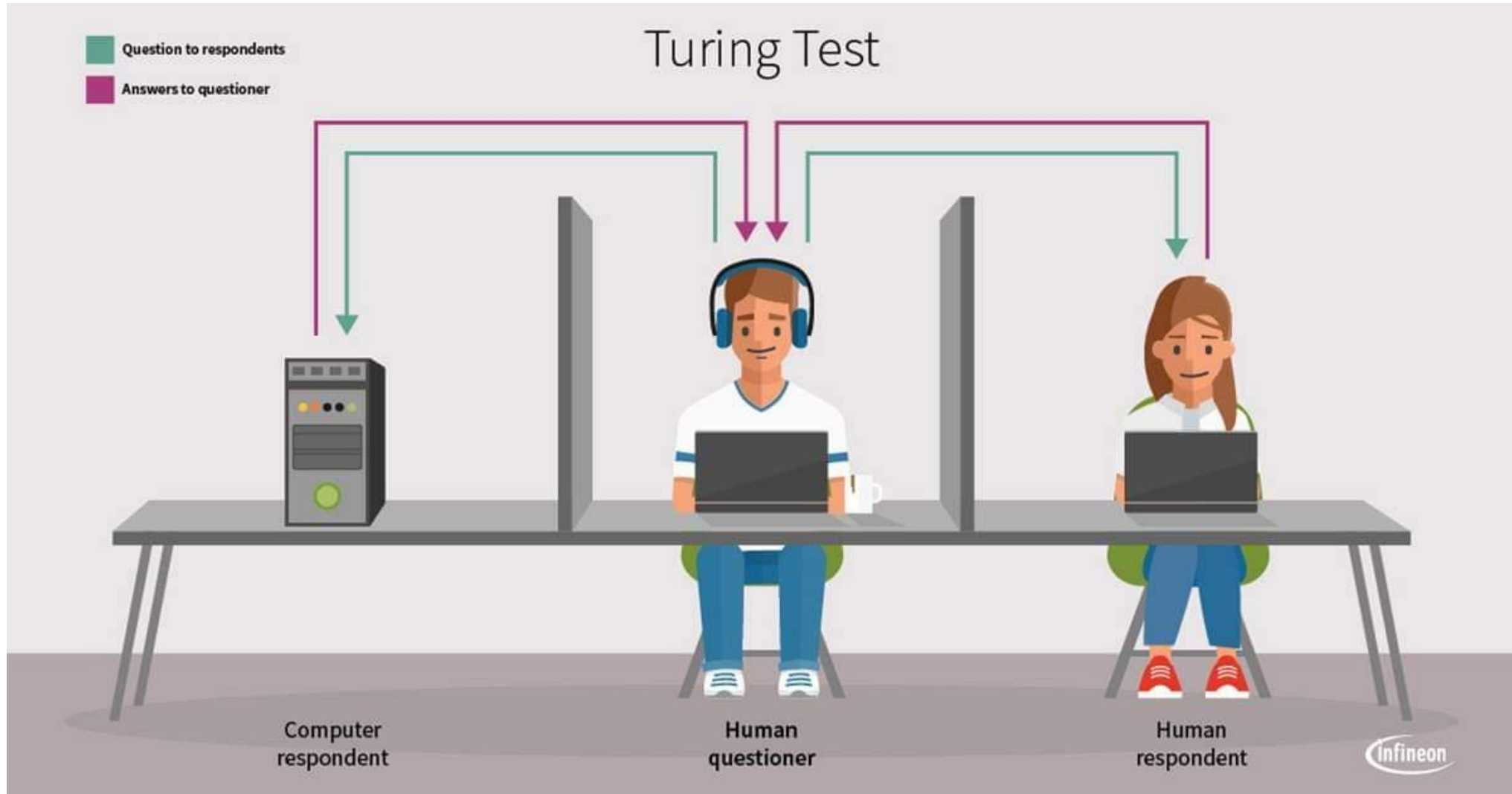
- Known as the **Rational Agent** approach.
 - AI systems developed with this approach are capable of acting rationally.

The Turing Test Approach

□ The Turing Test

- Proposed by **Alan Turing** in 1950.
- Also called the **imitation game**.
- Test a computer's ability to exhibit intelligent behavior equivalent to human.
- It involves a **computer**, a **human respondent** and a **human interrogator**.
- Direct physical interaction between them is avoided.
- The computer **passes the test** if the human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or a computer.

The Turing Test Approach



The Turing Test Approach

□ The Total Turing Test

- Proposed by **Stevan Harnad** in 1991.
- 2 additional requirements were added to create a stronger benchmark of intelligence:
 - **Perceptual ability**
 - **Object manipulation ability**

The Turing Test Approach

- The **pass the Total Turing Test**, a computer will have to possess at least the following 6 **capabilities**:
 - To communicate in human languages (**Natural language processing**)
 - To store what it knows or hears (**Knowledge representation**)
 - To do reasoning and to draw new conclusions (**Automated reasoning**)
 - To adapt to new circumstances and handle uncertainties (**Machine learning**)
 - To derive information from its percepts (**Computer vision**)
 - To manipulate objects and move about (**Robotics**)

- **Question:** Is the test still relevant today?

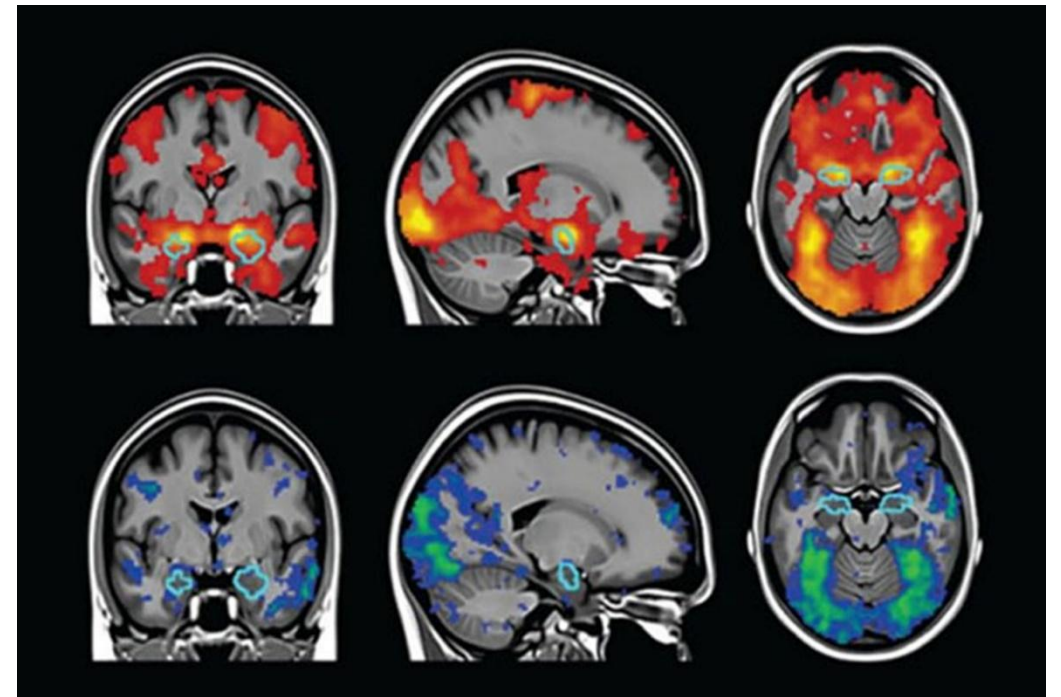
The Cognitive Modelling Approach

- A computer can **think like a human**.
- Need to understand **how humans think**.
- There are 3 ways to do this:
 - **Introspection**: Try to catch our own thoughts as they go by.



The Cognitive Modelling Approach

- There are 3 ways to do this:
 - **Psychological experiments:** Observe a person in action.
 - **Brain imaging:** Observe the brain in action.



The Cognitive Modelling Approach

□ Cognitive science

- An interdisciplinary field.
- Brings together computer models from A.I. and experimental techniques from psychology to construct precise and testable theories of the human mind.

The "Laws of Thought" Approach

□ Laws of Thought

- Rules that guide and underlie everyone's thinking, thoughts, expressions, discussions, etc.
- Serves as the underlying basis for rational discourse.
- Attempts to codify the "**right thinking**", or irrefutable **reasoning processes**.
- Initiated the field called **logic**.

The "Laws of Thought" Approach

□ Logic

- **Syllogism** is an example of laws of thought.
- It is a form of **deductive reasoning** consisting of two premises, and a conclusion.
- An **inference** is made when a logical conclusion is derived/drawn from premises known or assumed to be true.
- Example:

Sam is Michelle's parent. [Premise]

Sam is a woman. [Premise]

Therefore, Sam is Michelle's mother. [Conclusion]

The Rational Agent Approach

- An **agent** is something that acts.
- The term "agent" originates from the Latin's "agree", and means "to do something".
- A **rational agent** acts so as to **achieve the best outcome** or, when there is **uncertainty**, the **best expected outcome**.
- **Question:** What does **rationality** mean? (Chapter 2)

Which Approach?

□ Acting Rationally vs. Thinking Rationally

- One way to act rationally is to reason logically to the conclusion.
- However, correct inference doesn't necessarily result in rationality.
- Example:
 - Sometimes, no provably correct thing can be done, but something must still be done.
 - Reflex action that is usually more successful doesn't involve inference.
- Therefore, correct inference is just one of several possible mechanisms for achieving rationality.

Which Approach?

□ Acting Rationally vs. (Acting Humanly and Thinking Humanly)

- If the goal is to imitate human behaviors or thought processes, it can be challenging to model them mathematically, making the goal unachievable.
- However, the **standard of rationality** is **general**; yet **rationality** can be **well defined mathematically**, making it (provably) achievable.

□ Conclusion

- The **Rational Agent Approach** is **more practical** than other approaches.
- The view that **intelligence** is concerned mainly with **rational action** is adopted in this course.

Reading Homework

- **Artificial Intelligence: A Modern Approach**, 4th Edition, by Stuart Russell and Peter Norvig
 - Chapter 1: Introduction (page 19 to page 22)

The End